Data-Backed Journal-Style Replication of *Beyond Local Search*

Codex journal-replication workflow

2026-04-04

Abstract

This manuscript expands the local replication folder for Rosenkopf and Nerkar (2001) into a long journal-style paper using only processed datasets already present in the workspace. The current local batch contains 23,844 observations across 184 entities, and the batch headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. The draft imitates the structure of the source article while documenting the exact places where local replication decisions depart from direct source facts.

Keywords: boundary-spanning search; exploration; inventive impact; patent classes; local calculated analogue

1 Paper-First Reading Protocol

The workflow for this manuscript is intentionally ordered. The source paper is read first, the required empirical objects are identified second, the local files are verified third, and only then is the replication paper written from the checked evidence chain.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Ordered protocol used for this replication paper.

2 Required Data and Local File Manifest

This section answers the basic replication question of where the data are. Each empirical object is mapped to a concrete local file before the estimates are interpreted. Workspace root:

C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	No	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Yes	Executable local result file

Exact local files used by the manuscript.

3 Introduction

The source paper asks whether boundary-spanning search improves inventive impact relative to local search. The point of the present draft is not to pretend that every original source file has been recovered. It is to turn the executable local evidence into a paper that preserves theory, data architecture, variable mapping, and empirical claims in one place. The current local batch contains 23,844 observations across 184 entities, and the batch headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868. The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel.

4 Research Positioning and Contribution

The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different forms of exploratory search. The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. The writing stance follows a conservative paper-first workflow: the source paper defines the theoretical target, the local files define the executable boundary, and the replication paper names the distance between the two rather than smoothing it over.

5 Original Paper, Theory, and Published Estimand

The focal source article is [1]. The source paper is Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry by Lori Rosenkopf and Atul Nerkar in Strategic Management Journal . The source paper contrasts local search with boundary-spanning exploration. Impact rather than patenting volume is the main empirical outcome. Industry-specific coding is part of the original design. A useful replication discipline is to keep source facts and replication decisions separate. Source facts describe what the paper says. Replication decisions describe what the local executable workflow must do because it is limited to processed datasets already present in the workspace.

- The source paper contrasts local search with boundary-spanning exploration.
- Impact rather than patenting volume is the main empirical outcome.
- Industry-specific coding is part of the original design.

DirectSourceFact	LocalReplicationDecision
The source paper emphasizes different kinds of boundary-spanning search.	The local panel reconstructs a three-way search taxonomy from experience counts.
Impact is the key outcome.	The outcome remains impact-focused through citations and a top-tail indicator.
Industry-specific coding is part of the original design.	The industry-specific coding is generalized to the processed multi-firm patent panel.

Direct source facts are kept separate from local replication decisions.

6 Decision Register

The table below makes the major design choices explicit before any coefficient is interpreted.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether boundary-spanning search improves inventive impact relative to local search.
Unit of analysis	Patent

DecisionPoint	LocalSpecification
Sample frame	Patent-level recombination panel derived from the cleaned local reproduction dataset.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact and a yearly top-tail impact indicator.
Main explanatory design	Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.
Controls and fixed effects	Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.
Exclusion or substitution logic	The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.

Decision register for the local replication design.

7 Local Data Inventory and Sample Reconstruction

The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. Every number in this manuscript is backed by local files in patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search or by the explicitly referenced processed regression panel.

- Patent-level recombination panel built from the cleaned reproduction dataset.
- Search type reconstructed from prior organizational and field experience counts.
- Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator.

File	Role
rosenkopf_2001_analysis_dataset.csv	Patent-level search-type panel
rosenkopf_2001_model_coefficients.csv	Impact and top-5 coefficient table
rosenkopf_2001_sample_summary.csv	Search-type raw means

Local executable files used by this paper.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Sample reconstruction table from the local results folder.

8 Variable Mapping, Derivations, and Replication Decisions

The variable map is the bridge between the source article and the executable local workflow.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Local search	Prior organizational experience	search_type = local	Classify patents with positive prior organizational experience as local.
Boundary-existing-field search	No prior organizational but prior field experience	search_type = boundary_existing_field	Classify patents with zero prior organizational but positive field experience as boundary-existing-field.
Boundary-new-field search	No prior organizational and no prior field experience	search_type = boundary_new_field	Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.
Inventive impact	Patent-level citation impact	inventive_impact	Use the local patent-level impact variable from the cleaned panel.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Tail impact	Top-5-percent yearly impact indicator	high_top5_y	Use the already materialized yearly top-5-percent indicator.

Variable mapping from source constructs to local measures. Patent-level impact and top-tail models with search-type indicators and patent-level controls.

$$E[\text{Impact}_{pft} \mid X_{pft}] = \exp\left(\beta_1 \mathbb{1}[\text{BoundaryExisting}_p] + \beta_2 \mathbb{1}[\text{BoundaryNew}_p] + \beta_3 X_{pft} + \alpha_f + \gamma_t\right). \tag{1}$$

9 Empirical Design and Identification Limits

The local design remains strongest when it preserves the original unit of analysis and interaction logic, even if some measurement layers are reconstructed with defensible analogues rather than identical source files.

- The local search-type coding generalizes the original industry-specific implementation.
- Mismatch with published estimates can arise from both sample expansion and measurement differences.
- The exercise is most informative about directional mechanisms.

10 Main Results

Boundary existing field versus local is estimated at -0.0389 with p = 0.3868 in model rosenkopf_impact. Boundary new field versus local is estimated at 0.1126 with p = 0.0014 in model rosenkopf_impact.

Breadth of recombination is estimated at 0.0223 with p = 0.0870 in model rosenkopf_impact. Recombination degree is estimated at -0.4108 with p = 0.0226 in model rosenkopf_impact. Prior-art citations is estimated at 0.0013 with p = 0.0548 in model rosenkopf_impact. Log prior patent stock is estimated at -0.1965 with p = 0.0657 in model rosenkopf_impact. Boundary existing field versus local is estimated at -0.2317 with p = 0.1534 in model rosenkopf_top5. Boundary new field versus local is estimated at 0.1039 with p = 0.3585 in model rosenkopf_top5. Breadth of recombination is estimated at 0.0069 with p = 0.7349 in model rosenkopf_top5. Recombination degree is estimated at -0.0597 with p = 0.8303 in model rosenkopf_top5. Prior-art citations is estimated at 0.0055 with p = 0.1746 in model rosenkopf_top5. Log prior patent stock is estimated at -0.4208 with p = 0.0184 in model rosenkopf_top5.

model	Term	estimate	std_error	p_value
rosenkopf_impact	Boundary existing field versus local	-0.0389	0.0449	0.3868
rosenkopf_impact	Boundary new field versus local	0.1126	0.0353	0.0014
rosenkopf_impact	Breadth of recombination	0.0223	0.0130	0.0870
rosenkopf_impact	Recombination degree	-0.4108	0.1802	0.0226
rosenkopf_impact	Prior-art citations	0.0013	0.0007	0.0548
rosenkopf_impact	Log prior patent stock	-0.1965	0.1067	0.0657
rosenkopf_top5	Boundary existing field versus local	-0.2317	0.1623	0.1534
rosenkopf_top5	Boundary new field versus local	0.1039	0.1131	0.3585
rosenkopf_top5	Breadth of recombination	0.0069	0.0205	0.7349

model	Term	estimate	std_error	p_value
rosenkopf_top5	Recombination degree	-0.0597	0.2786	0.8303
rosenkopf_top5	Prior-art citations	0.0055	0.0041	0.1746
rosenkopf_top5	Log prior patent stock	-0.4208	0.1786	0.0184

Key coefficient table used in the main text.

Term	Model	Estimate	PValue	Signal	Reading
Boundary existing field versus local	rosenkopf_impact	-0.0389	0.3868	Weak	Negative estimate with p = 0.3868.
Boundary new field versus local	rosenkopf_impact	0.1126	0.0014	Strong	Positive estimate with p = 0.0014.
Breadth of recombination	rosenkopf_impact	0.0223	0.0870	Suggestive	Positive estimate with p = 0.0870.
Recombination degree	rosenkopf_impact	-0.4108	0.0226	Moderate	Negative estimate with p = 0.0226.

Term	Model	Estimate	PValue	Signal	Reading
Prior-art citations	rosenkopf_impact	0.0013	0.0548	Suggestive	Positive estimate with p = 0.0548.
Log prior patent stock	rosenkopf_impact	-0.1965	0.0657	Suggestive	Negative estimate with p = 0.0657.
Boundary existing field versus local	rosenkopf_top5	-0.2317	0.1534	Weak	Negative estimate with p = 0.1534.
Boundary new field versus local	rosenkopf_top5	0.1039	0.3585	Weak	Positive estimate with p = 0.3585.
Breadth of recombination	rosenkopf_top5	0.0069	0.7349	Weak	Positive estimate with p = 0.7349.
Recombination degree	rosenkopf_top5	-0.0597	0.8303	Weak	Negative estimate with p = 0.8303.
Prior-art citations	rosenkopf_top5	0.0055	0.1746	Weak	Positive estimate with p = 0.1746.

Term	Model	Estimate	PValue	Signal	Reading
Log prior patent stock	rosenkopf_top5	-0.4208	0.0184	Moderate	Negative estimate with p = 0.0184.

Interpretive summary of the main coefficients.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	23844.0000	5.3099	17.5532	1.0000	2.0000	6.0000	0.0000	812.00
Top-5 indicator	23844.0000	0.0411	0.1985	0.0000	0.0000	0.0000	0.0000	1.00
Breadth of recombination	23844.0000	4.2941	4.5669	2.0000	3.0000	5.0000	2.0000	117.00
Recombination degree	23844.0000	0.5549	0.1738	0.4444	0.5600	0.6914	0.0454	0.95
Field experience	23844.0000	1.6339	1.8203	0.0000	1.0986	2.7726	0.0000	8.01
Organizational experience	23844.0000	0.9227	1.3174	0.0000	0.6931	1.3863	0.0000	7.65
Prior-art citations	23844.0000	15.0909	50.4465	5.0000	8.0000	13.0000	2.0000	2263.00
Same-year patents	23844.0000	728.7229	1302.8291	109.0000	313.0000	652.0000	1.0000	7321.00
Prior cumulative stock	23844.0000	7.5074	2.2142	6.3297	7.6930	8.8934	0.0000	11.32

Descriptive statistics for the main variables.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	13675.0000	2.4112	4.4643
2.0000	3458.0000	4.0000	4.9251
3.0000	2273.0000	5.0000	5.5055
4.0000	4438.0000	9.9635	8.1151

Quantile summary linking the main regressor to the local outcome.

11 Claim-Evidence Audit

This audit prevents the paper from blurring together original-paper statements, local estimates, and replication judgments.

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether boundary-spanning search improves inventive impact relative to local search.	Direct source fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	Recovered from the source article before local writing begins.
The local executable panel contains 23,844 observations across 184 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Backed by the verified analysis panel and sample audit.

Claim	EvidenceType	EvidenceFile	AuditNote
The local headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Backed by the exported coefficient file used in the paper.
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Claim-evidence audit for the current manuscript.

12 Comparison to the Published Paper and Remaining Gaps

The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise. The honest replication standard is to describe the gap without smoothing it over. A useful local paper tells the next researcher exactly which parts of the original design are recoverable now and which parts remain approximated.

13 Discussion and Conclusion

The manuscript turns a replication folder into a paper that can be uploaded, cited internally, and rerun later without guessing how the evidence was assembled. For Rosenkopf and Nerkar (2001)

the bottom line is: The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel.

A Appendix A: Source Inventory

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	No	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv	Yes	Executable local dataset

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

B Appendix B: Reading Protocol

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

C Appendix C: Decision Register

DecisionPoint	LocalSpecification
Research question	The source paper asks whether boundary-spanning search improves inventive impact relative to local search.
Unit of analysis	Patent
Sample frame	Patent-level recombination panel derived from the cleaned local reproduction dataset.
Time window	1986 to 2016 in the local executable panel.
Outcome definition	Patent-level inventive impact and a yearly top-tail impact indicator.
Main explanatory design	Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search.
Controls and fixed effects	Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Detailed sample reconstruction table.

E Appendix E: Variable Mapping

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Local search	Prior organizational experience	search_type = local	Classify patents with positive prior organizational experience as local.
Boundary-existing-field search	No prior organizational but prior field experience	search_type = boundary_existing_field	Classify patents with zero prior organizational but positive field experience as boundary-existing-field.
Boundary-new-field search	No prior organizational and no prior field experience	search_type = boundary_new_field	Classify patents with zero prior organizational and zero prior field experience as boundary-new-field.
Inventive impact	Patent-level citation impact	inventive_impact	Use the local patent-level impact variable from the cleaned panel.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Tail impact	Top-5-percent yearly impact indicator	high_top5_y	Use the already materialized yearly top-5-percent indicator.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	23844.0000	5.3099	17.5532	1.0000	2.0000	6.0000	0.0000	812.00
Top-5 indicator	23844.0000	0.0411	0.1985	0.0000	0.0000	0.0000	0.0000	1.00
Breadth of recombination	23844.0000	4.2941	4.5669	2.0000	3.0000	5.0000	2.0000	117.00
Recombination degree	23844.0000	0.5549	0.1738	0.4444	0.5600	0.6914	0.0454	0.95
Field experience	23844.0000	1.6339	1.8203	0.0000	1.0986	2.7726	0.0000	8.01
Organizational experience	23844.0000	0.9227	1.3174	0.0000	0.6931	1.3863	0.0000	7.65
Prior-art citations	23844.0000	15.0909	50.4465	5.0000	8.0000	13.0000	2.0000	2263.00
Same-year patents	23844.0000	728.7229	1302.8291	109.0000	313.0000	652.0000	1.0000	7321.00
Prior cumulative stock	23844.0000	7.5074	2.2142	6.3297	7.6930	8.8934	0.0000	11.32

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Inventive impact	Top-5 indicator	Breadth of recombination	Recombination degree	Field experience	Organizational experience	Prior-art citations	Same-year patents	Prior cumulative stock
Inventive impact	1.000	0.455	0.364	0.018	-0.021	-0.008	0.591	-0.018	0.022
Top-5 indicator	0.455	1.000	0.129	0.024	-0.012	0.006	0.149	-0.015	0.005
Breadth of recombination	0.364	0.129	1.000	0.466	-0.263	-0.144	0.822	0.011	0.149
Recombination degree	0.018	0.024	0.466	1.000	-0.456	-0.313	0.140	0.065	0.122
Field experience	-0.021	-0.012	-0.263	-0.456	1.000	0.838	-0.077	0.060	0.132
Organizational experience	-0.008	0.006	-0.144	-0.313	0.838	1.000	-0.018	0.152	0.267
Prior-art citations	0.591	0.149	0.822	0.140	-0.077	-0.018	1.000	-0.016	0.081
Same-year patents	-0.018	-0.015	0.011	0.065	0.060	0.152	-0.016	1.000	0.520
Prior cumulative stock	0.022	0.005	0.149	0.122	0.132	0.267	0.081	0.520	1.000

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

Period	V1	V2	V3	V4	V5
1986.0000	643.0000	4.0202	2.8694	0.5175	0.4277
1987.0000	789.0000	4.2801	2.9176	0.5177	0.4335
1988.0000	1066.0000	4.4606	3.1323	0.5319	0.4353
1989.0000	1390.0000	3.9338	3.0813	0.5272	0.4007
1990.0000	1349.0000	4.5137	3.1156	0.5218	0.3744
1991.0000	1325.0000	4.7381	3.1502	0.5314	0.3743
1992.0000	1302.0000	5.0154	3.3318	0.5346	0.4040
1993.0000	1354.0000	5.6721	3.4609	0.5323	0.3877
1994.0000	1330.0000	6.2654	3.5782	0.5413	0.3932
1995.0000	1217.0000	6.2597	3.8439	0.5563	0.3780
1996.0000	659.0000	6.7269	3.8376	0.5306	0.3763
1997.0000	646.0000	6.0139	4.2446	0.5521	0.4195
1998.0000	356.0000	6.2949	3.9242	0.5473	0.4045
1999.0000	654.0000	5.4832	3.8104	0.5436	0.3700
2000.0000	636.0000	5.5409	4.0220	0.5402	0.3632
2001.0000	614.0000	5.1515	4.4137	0.5625	0.4169
2002.0000	632.0000	4.8892	4.3845	0.5578	0.3687
2003.0000	522.0000	4.5019	4.4176	0.5505	0.3487
2004.0000	324.0000	3.6636	4.8086	0.5734	0.4136
2005.0000	284.0000	4.5000	5.4859	0.5763	0.4085
2006.0000	336.0000	4.8899	5.2649	0.5661	0.3214
2007.0000	246.0000	3.6545	5.2683	0.5692	0.3089
2008.0000	535.0000	3.6523	4.5121	0.5904	0.3607
2009.0000	206.0000	4.8010	6.0049	0.5992	0.2961
2010.0000	511.0000	4.4168	6.6477	0.6078	0.2564
2011.0000	651.0000	6.4516	6.2750	0.6090	0.2550

Period	V1	V2	V3	V4	V5
2012.0000	1068.0000	4.0000	6.5187	0.6113	0.2603
2013.0000	868.0000	5.3859	6.7304	0.5978	0.2604
2014.0000	725.0000	12.6428	6.9890	0.6000	0.2290
2015.0000	1072.0000	6.6157	6.1744	0.5971	0.2043
2016.0000	534.0000	3.7678	4.6292	0.5811	0.1667

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Quantile	N	MeanRegressor	MeanOutcome
1.0000	13675.0000	2.4112	4.4643
2.0000	3458.0000	4.0000	4.9251
3.0000	2273.0000	5.0000	5.5055
4.0000	4438.0000	9.9635	8.1151

Quantile summary for the main regressor.

SearchType	Patents	MeanImpact	Top5Share
boundary_new_field	8443.0000	5.8602	0.0458
local	12131.0000	5.1357	0.0411
boundary_existing_field	3270.0000	4.5352	0.0287

Raw group means from the local summary file.

J Appendix J: Full Coefficient Table

term	estimate	std_error	statistic	p_value	model
search_type::boundary_existing_field	-0.0389	0.0449	-0.8655	0.3868	rosenkopf_impact
search_type::boundary_new_field	0.1126	0.0353	3.1882	0.0014	rosenkopf_impact
breadth_recombination	0.0223	0.0130	1.7114	0.0870	rosenkopf_impact
recombination_degree	-0.4108	0.1802	-2.2800	0.0226	rosenkopf_impact
n_prior_art_cites	0.0013	0.0007	1.9206	0.0548	rosenkopf_impact
total_patents_same_year	-0.0001	0.0000	-2.0994	0.0358	rosenkopf_impact
total_patents_until_prev_log	-0.1965	0.1067	-1.8408	0.0657	rosenkopf_impact
search_type::boundary_existing_field	-0.2317	0.1623	-1.4276	0.1534	rosenkopf_top5
search_type::boundary_new_field	0.1039	0.1131	0.9183	0.3585	rosenkopf_top5
breadth_recombination	0.0069	0.0205	0.3386	0.7349	rosenkopf_top5
recombination_degree	-0.0597	0.2786	-0.2143	0.8303	rosenkopf_top5
n_prior_art_cites	0.0055	0.0041	1.3577	0.1746	rosenkopf_top5
total_patents_same_year	-0.0001	0.0001	-0.9622	0.3360	rosenkopf_top5
total_patents_until_prev_log	-0.4208	0.1786	-2.3566	0.0184	rosenkopf_top5

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Inventive impact	0.0000	0.0000
Top-5 indicator	0.0000	0.0000
Breadth of recombination	0.0000	0.0000
Recombination degree	0.0000	0.0000
Field experience	0.0000	0.0000
Organizational experience	0.0000	0.0000
Prior-art citations	0.0000	0.0000
Same-year patents	0.0000	0.0000
Prior cumulative stock	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
6066.0000	3200.0000	1986.0000	2016.0000
5047.0000	1318.0000	1986.0000	2016.0000
4194.0000	1017.0000	1986.0000	2010.0000
7435.0000	854.0000	1986.0000	2016.0000
4087.0000	830.0000	1986.0000	2016.0000
4839.0000	815.0000	1986.0000	2016.0000
7585.0000	774.0000	1986.0000	2016.0000
10499.0000	755.0000	1986.0000	2016.0000
5073.0000	671.0000	1986.0000	2008.0000
4060.0000	574.0000	1986.0000	2016.0000
1300.0000	551.0000	1986.0000	2016.0000
8762.0000	524.0000	1986.0000	2016.0000
1690.0000	452.0000	1990.0000	2016.0000
1581.0000	421.0000	1986.0000	2004.0000
10983.0000	403.0000	1986.0000	2016.0000
12141.0000	398.0000	1995.0000	2016.0000
4503.0000	344.0000	1986.0000	2016.0000
6266.0000	340.0000	1986.0000	2016.0000
2285.0000	334.0000	1986.0000	2016.0000
7475.0000	300.0000	1986.0000	1998.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether boundary-spanning search improves inventive impact relative to local search.	Direct source fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf	Recovered from the source article before local writing begins.

Claim	EvidenceType	EvidenceFile	AuditNote
The local executable panel contains 23,844 observations across 184 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv	Backed by the verified analysis panel and sample audit.
The local headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868.	Local calculated fact	patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv	Backed by the exported coefficient file used in the paper.
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Source Inventory Note

Source Inventory - paper: Rosenkopf and Nerkar (2001) - source title: Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry - source file: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf‘ ## Paper-First Reading Protocol - Step 1: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf‘ — needed for: Read the original paper before touching the local estimates. — exists: n/a — status: workflow step - Step 2: ‘Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry‘ — needed for: Extract the source paper’s theoretical logic, sample definition, variable definitions, and estimand. — exists: n/a — status: workflow step - Step 3: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv‘ — needed for: Match each required empirical object to a local executable file and verify that the file exists. — exists: n/a — status: workflow step - Step 4: ‘The manuscript, markdown notes, and PDF are generated from the verified file set.‘ — needed for: Write the replication paper only from the checked local files and calculated outputs. — exists: n/a — status: workflow step ## Local files - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: No — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive -

Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coeffic — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_20 — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

O Appendix O: Study Map Note

Study Map - research question: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. - positioning: The replication matters because boundary-spanning search claims are often treated broadly, while the original paper distinguishes among qualitatively different forms of exploratory search. - contribution: The paper's contribution is a transparent three-way search taxonomy on a local executable panel that separates local, boundary-existing-field, and boundary-new-field search. - local design: The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. - estimand: Patent-level impact and top-tail models with search-type indicators and patent-level controls. - headline estimate: '-0.0389' with 'p = 0.3868'. ## Decision Audit - Research question: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. - Unit of analysis: Patent - Sample frame: Patent-level recombination panel derived from the cleaned local reproduction dataset. - Time window: 1986 to 2016 in the local executable panel. - Outcome definition: Patent-level inventive impact and a yearly top-tail impact indicator. - Main explanatory design: Patent-level impact models with search-type indicators comparing boundary-spanning forms to local search. - Controls and fixed effects: Patent-level controls for breadth, degree, citations, scale, and prior stock with available fixed effects. - Exclusion or substitution logic: The original industry-specific coding is not locally recoverable, so the executable rerun generalizes the taxonomy using prior organizational and field experience counts. ## Source Versus Decision - direct source fact: The source paper emphasizes different kinds of boundary-spanning search. - direct source fact: Impact is the key outcome. - direct source fact: Industry-specific coding is part of the original design. local replication decision: The local panel reconstructs a three-way search taxonomy from experience counts. local replication decision: The outcome remains impact-focused through citations and a top-tail indicator. local replication decision: The industry-specific coding is generalized to the processed multi-firm patent panel.

P Appendix P: Data Design Note

Data Design - Patent-level recombination panel built from the cleaned reproduction dataset. - Search type reconstructed from prior organizational and field experience counts. - Impact outcomes measured with patent-level inventive impact and a top-5-percent indicator. - panel file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv‘ - coefficient file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients. - sample summary used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_ - observations: ‘23,844‘ - entities: ‘184‘ ## Required Data And Local File Mapping - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: No — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coeffic — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_20 — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

Q Appendix Q: Derivations Note

Derivations And Estimands - The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. - Patent-level impact and top-tail models with search-type indicators and patent-level controls. - model equations are reproduced in ‘paper/main.tex‘. ## Variable Map - Local search -¿ ‘search_type = local‘ : Classify patents with positive prior organizational experience as local. - Boundary-existing-field search -¿ ‘search_type = boundary_existing_field‘ : Classify patents with zero prior organizational but positive field experience as boundary-existing-field. - Boundary-new-field search -¿ ‘search_type = boundary_new_field‘ : Classify patents with zero prior organizational and zero prior field experience as boundary-new-field. - Inventive impact -¿ ‘inventive_impact‘ : Use the local patent-level impact variable from the cleaned panel. - Tail impact -¿ ‘high_top5_y‘ : Use the already materialized yearly top-5-percent indicator. ## Claim Evidence Audit - Direct source fact: The source paper studies: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. — evidence: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/patent literature/Strategic Management Journal - 2001 - Rosenkopf - Beyond local search boundary-spanning exploration and impact in the.pdf - Local calculated fact: The local executable panel contains 23,844 observations across 184 entities. — evidence: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_sample_summary.csv - Local calculated fact: The local headline estimate is -0.0389 for boundary_existing_field vs local with p = 0.3868. — evidence: patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_model_coefficients.csv - Replication decision: The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable. — evidence: Source-vs-decision table and limitations section

R Appendix R: Replication Decisions Note

Replication Decisions - The local search-type coding generalizes the original industry-specific implementation. - Mismatch with published estimates can arise from both sample expansion and measurement differences. - The exercise is most informative about directional mechanisms. - direct source facts are separated from local replication decisions in the manuscript. - no coefficient or table is fabricated. ## Source Facts Versus Local Decisions - source: The source paper emphasizes different kinds of boundary-spanning search. - source: Impact is the key outcome. - source: Industry-specific coding is part of the original design. decision: The local panel reconstructs a three-way search taxonomy from experience counts. decision: The outcome remains impact-focused through citations and a top-tail indicator. decision: The industry-specific coding is generalized to the processed multi-firm patent panel.

S Appendix S: Local Execution Status Note

Local Execution Status - batch date: ‘2026-04-04‘ - status: long-form journal-style manuscript generated from local result files - current paper: ‘rosenkopf-2001-beyond-local-search‘

T Appendix T: Cleaned Output Tables Note

Cleaned Output Tables - paper-first reading protocol - required-data and file-manifest table - decision register - sample reconstruction table - local source inventory table - source facts versus replication decisions table - claim-evidence audit table - variable mapping table - key coefficient table - full coefficient table - descriptive statistics - correlation matrix - yearly audit - missingness audit - top-entity coverage table - quantile summary

U Appendix U: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

V Appendix V: Manuscript Metadata

Field	Value
Paper	Rosenkopf and Nerkar (2001)
Source title	Beyond Local Search: Boundary-Spanning, Exploration, and Impact in the Optical Disk Industry
Authors	Lori Rosenkopf and Atul Nerkar
Outlet	Strategic Management Journal
Batch date	2026-04-04
Rows	23,844
Entities	184
Headline term	boundary_existing_field vs local
Headline estimate	-0.0389
Headline p-value	0.3868
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/rosenkopf-2001-beyond-local-search/results/rosenkopf_2001_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

W Appendix W: Replication Summary Note

Research question: The source paper asks whether boundary-spanning search improves inventive impact relative to local search. Local design: The local analogue reconstructs local, boundary-existing-field, and boundary-new-field search from organizational and field experience counts already materialized in the processed recombination panel. Estimand: Patent-level impact and top-tail models with search-type indicators and patent-level controls. Local result focus: The local analogue supports the most novel boundary-spanning channel much more clearly than the moderate boundary-spanning channel. Comparison note: The local evidence is strongest for boundary-new-field search rather than for boundary-existing-field search. Boundary-new-field patents receive higher average impact relative to local search, while boundary-existing-field patents are lower and imprecise.

References

[1] Lori Rosenkopf and Atul Nerkar. Beyond local search: Boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4):287–306, 2001.